

Table 1. Evolution of the universe since the Big Bang in million years (M) and million years ago (Ma) by scale (as of May 2026).

Ma	M	Universal Scale		Large Scale	Stellar Scale
13800	0	0.38-400 M: Dark Ages	~150-1000 M: Hydrogen Reionization	150 M: Cosmic dawn	
13600	200	~150-300 M: Dark Matter			
13400	400				
13200	600				
13000	800				
12800	1000				
12600	1200				
12400	1400				
12200	1600				
12000	1800				
11800	2000		~2000-3000 M: Helium II Reionization	~2000-3000 M: Quasar era peak	
11600	2200			2000-3000 M: Cosmic noon	~2500 M: Peak star birth
11400	2400				
11200	2600				
11000	2800				
10800	3000			3000-5000 M: Cosmic afternoon	
10600	3200				
10400	3400				
10200	3600				
10000	3800			~10000 Ma: Quasar era decline	
9800	4000				
9600	4200				
9400	4400				
9200	4600				
9000	4800				
8800	5000			~5000 M: Milky Way, disk shape ¹	~5000 M: Star birth starting to decline
8600	5200				
8400	5400				
8200	5600				
8000	5800			~6000 M: Galaxy Cluster Maturation ²	
7800	6000				
7600	6200				
7400	6400				
7200	6600				
7000	6800				
6800	7000				~7000 M: Milky Way ³
6600	7200				
6400	7400	~5000-6000 Ma: Rise of Dark Energy ⁴			
6200	7600				
6000	7800				
5800	8000				
5600	8200				
5400	8400				
5200	8600				
5000	8800				
4800	9000				
4600	9200				~4600 Ma: Sun, Earth, Moon
4400	9400	~4000 Ma: Energy Density Dominance ⁵			
4200	9600				
4000	9800				
3800	10000				
3600	10200				
3400	10400				
3200	10600				
3000	10800				
2800	11000				~3850 Ma: No O ₂ in the atm. (PRE-GOE)
2600	11200				
2400	11400				~3000 Ma: Cyanobacteria ⁶
2200	11600				
2000	11800				
1800	12000				
1600	12200				
1400	12400				
1200	12600				
1000	12800				
800	13000				
600	13200				~800–700 Ma: First Animals
400	13400				~630–600 Ma: First bilateral symmetry
200	13600				~541–485 Ma: Cambrian Explosion ⁷
0	13800				~251–201 Ma: Great Dying extinction ⁸
					66 Ma: Mass extinction, Cenozoic Era ⁹

Note:
1) The spinning gas and stars settled into a distinct, thin spiral disk.
2) During this era, large-scale structures like galaxy superclusters and filaments continued to stabilize, though the accelerating expansion began to make it harder for new, massive structures to gravitationally bind together.
3) The Milky Way is beginning to take its final shape.
4) Dark Energy was just beginning to become the dominant force over gravity, starting to accelerate the expansion of the universe after several billion years of slowing down.
5) Dark Energy's negative pressure is highly efficient.
6) First to perform photosynthesis, leading to the 'Great Oxygenation Event'.
7) A massive burst of evolution occurs, establishing nearly all modern animal phyla, including mollusks, arthropods, and chordates.
8) Triassic Era, after the 'Great Dying' extinction (the Permian–Triassic extinction event), the first dinosaurs and early mammals evolve.
9) The modern Era, a 'mass extinction' (the Cretaceous–Paleogene extinction event K–Pg) wipes out non-avian dinosaurs, allowing mammals to rapidly diversify and become dominant.